MojoBrain

Architecture Guide

How to Build a 7-Agent AI Operating System

Mission Control — your live agent dashboard

eyshtech.vip | By Eysha Penrod

The Problem

You built a business. Now it's running you.

Every week you spend 15–20 hours on operations that have nothing to do with serving clients — follow-ups you forgot to send, invoices you haven't checked, that security setting you've been meaning to audit, the lead that replied three days ago and is now cold. You know the list. It haunts you on Sunday nights.

I know because I was you. I spent 6 years as a Google Workspace Sr. Account Manager at MarketStar, managing environments for 250+ SMBs. I watched the same pattern everywhere: smart business owners drowning in admin work that should run itself.

When I went out on my own, it got worse. I was the CEO, the IT department, the sales team, and the compliance officer — all at once. I almost hired a VA. Then I realized I'd just be managing the VA.

> 15-20 hours/week on ops x $100/hr opportunity cost
> = $6,000-8,000/month in lost revenue.
> MojoBrain costs ~$5-20/mo in API fees to run.

So I built MojoBrain. Seven autonomous AI agents that handle lead gen, outreach, compliance, finances, and security. Running 24/7 on a single Mac Mini. I approve from my phone every morning. The agents handle the rest.

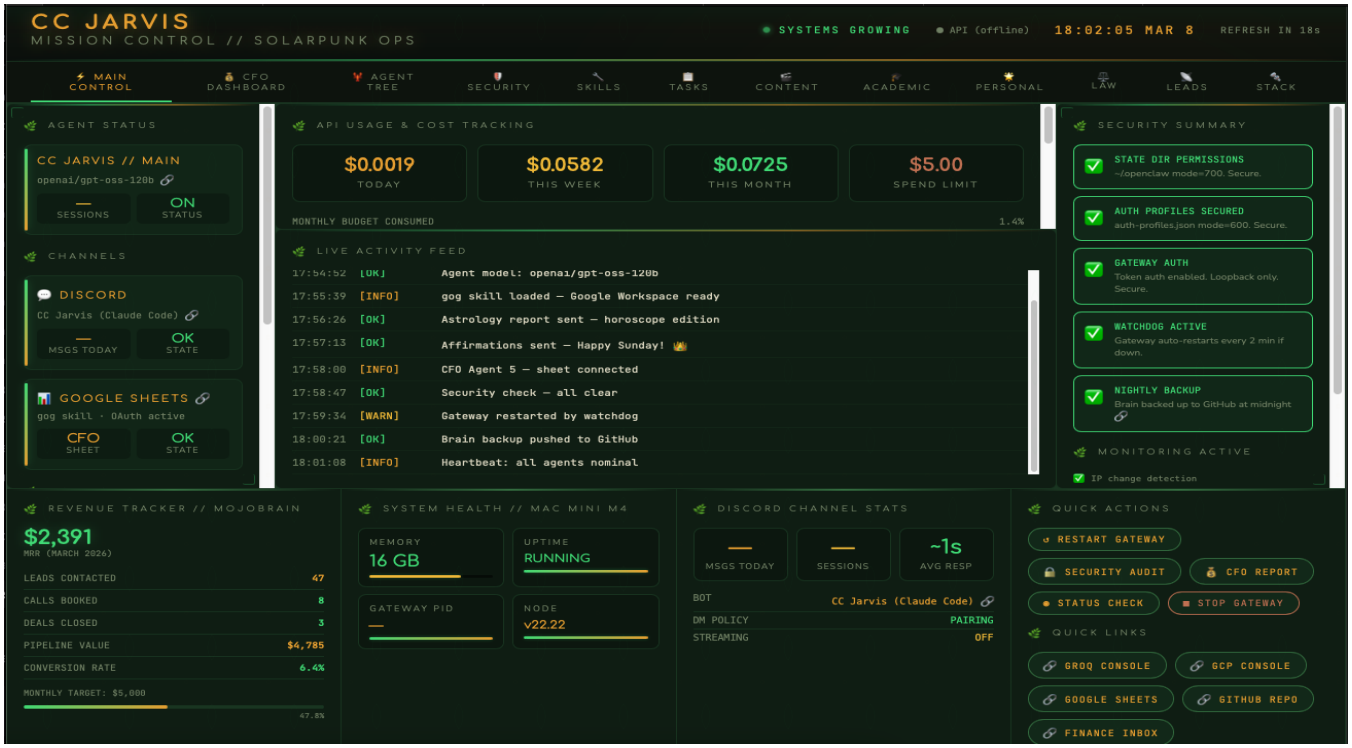
I named it Mojo. It felt right.

What Actually Runs

Let me walk you through it. MojoBrain is a collection of Python scripts that run on cron schedules — every 5 minutes, every 15 minutes, every hour, depending on the agent. Each one writes to a JSON state file. The Nerve Center reads all of those outputs, builds a unified picture, and routes events between agents when something cross-functional happens.

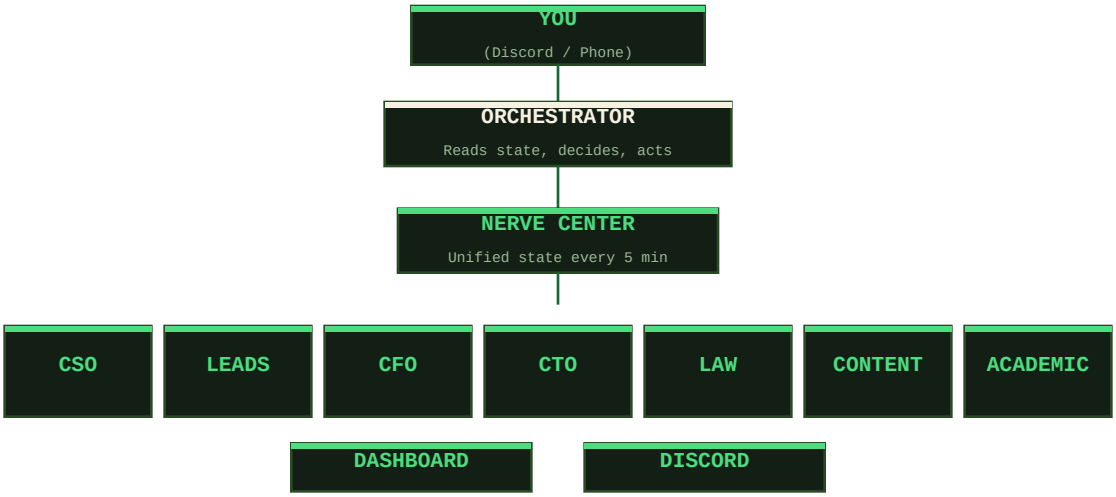
You get notifications via Discord. Critical stuff goes to one channel. Routine logs go to another. You see what matters. Nothing more.

MISSION CONTROL — LIVE VIEW OF ALL AGENTS



Real-time agent status, financial tracking, security alerts, system health. One screen.

SYSTEM ARCHITECTURE


YOU
(Discord / Phone)
ORCHESTRATOR
Reads state, decides, acts
NERVE CENTER
Unified state every 5 min
CSO
LEADS
CFO
CTO
LAW
CONTENT
ACADEMIC
DASHBOARD
DISCORD

Meet the Team

Each agent runs autonomously. No babysitting. Think of them less like software and more like employees who never sleep, never forget, and never complain.

Orchestrator
CENTRAL INTELLIGENCE

The brain. Reads all agent outputs, makes decisions, routes events. It's the one that sends you your morning briefing and decides what deserves your attention.

CSO
SECURITY

Paranoid in the best way. Checks DNS records, scans for breach mentions, audits SSL certs — every 15 minutes. I haven't thought about email security since I deployed it.

Lead Engine
LEAD GENERATION

This is the one that paid for itself first. Scrapes, validates, scores, and queues leads while you sleep. Replaced 15 hours of manual prospecting per week.

Nerve Center
COORDINATION

The nervous system. Reads ALL agent outputs every 5 minutes, builds a unified state, detects patterns. This is what makes it a system, not just scripts.

CFO
FINANCE

Tracks your burn rate, monitors subscriptions, flags suspicious charges, knows your tax deadlines. Your financial watchdog that never takes a day off.

CTO
INFRASTRUCTURE

Monitors system health, disk space, cron jobs, and runs automated recovery when things go sideways. Because things always go sideways at 3am.

Law
COMPLIANCE

GDPR scanning, disclaimer tracking, contract clause library, compliance routing. Keeps you covered without a $500/hr retainer.

What This Costs to Run

Here's exactly what this costs to run. I was shocked when I added it up the first time.

~$20
Monthly API Cost

7
Autonomous Agents

24/7
Unattended Operation

Component	Technology	Cost
Hardware	Mac Mini / Linux / VM	$0 (use yours)
LLM	Your choice (Claude, GPT, Groq)	$5-20/mo
Orchestration	Python 3 + cron	$0
Dashboard	Self-contained HTML	$0
Notifications	Discord webhooks	$0
Email	Brevo (5K emails/mo)	$9/mo
Domain	Namecheap	$7/yr
Hosting	GitHub Pages	$0

The $0 cells aren't a trick. Groq gives you a genuinely useful free tier. Discord is free forever. GitHub is free for private repos. The only real cost is your LLM provider — and that's the brain that makes everything else work.

How I Got Here

MojoBrain didn’t start as a product. It started as survival.

I was running a consulting business while employed full-time, managing tools manually, and losing leads because I couldn’t follow up fast enough. The gateway I’d built crashed at 3am on a Tuesday and took down everything. That was the moment I decided to rebuild from scratch.

> The first agent I built was the Lead Engine.
> Night one, it scraped and scored 47 leads while I slept.
> That one agent replaced 15 hours of manual prospecting per week.

Within two weeks I had 7 agents running. The system evolved from a collection of scripts into something that felt alive — a coordinated operating system with a central nervous system, a dashboard, and autonomous decision-making.

> The breakthrough wasn't any single agent.
> It was the Nerve Center — the component that reads ALL agent outputs
> every 5 minutes and coordinates cross-agent responses.
> That's when it stopped being scripts and became a system.

I’m telling you this because if your system is currently broken, the answer is probably not to fix it. It’s to replace it with something simpler.

Start Here

The first agent I'd build if I were starting over is the security monitor. Not because it's the flashiest — it's not. But because it runs silently in the background and one day it's going to catch something that would have cost you real money or real embarrassment.

01 — System Health Monitor

A Python script that checks disk space, memory, and whether your cron jobs actually ran. Every 15 minutes it writes a status file. If anything is wrong, it pings you on Discord. Takes about 30 minutes to build. You'll wonder how you lived without it.

```
Stack: Python + psutil + Discord webhook
  Schedule: */15 * * * *
  Build time: 30 minutes
```

02 — Morning Briefing Bot

This one changed my mornings. It pulls data from your tools — email count, calendar events, task list, whatever you care about — and sends you a formatted summary at 7am. Like having a chief of staff who works for free and never calls in sick.

```
Stack: Python + cron + API calls
  Schedule: 0 11 * * * (7am ET)
  Build time: 1 hour
```

03 — Lead Scraper + Scorer

This is the one that makes money. It searches for businesses matching your ideal customer, validates their domains (no point pitching a parked domain), and scores them on signals like hiring posts, tech stack, and company size. By morning you have a ranked list of people who actually need what you sell.

```
Stack: Python + DuckDuckGo + DNS checks
  Schedule: 0 2 * * * (2am)
  Build time: 2-3 hours
```

Where to Go From Here

If you’ve read this far, you’re probably one of two people.

Either you’re going to build this yourself — great, the patterns are all here. The three agents above are a real starting point, not a teaser. Build them this weekend and you’ll have a system that’s already better than what 90% of businesses are running.

Or you’re thinking “I want this but I don’t want to build it.”

That’s what I do. Three ways to work together:

Playbook — $297	All 7 agent scripts, Mission Control dashboard, deployment playbook with video walkthroughs.
Guided Setup — $797	Everything in Playbook plus a 2-hour live setup call, your Lead Engine customized to your mar
Managed Deploy — $1,497	I install everything on your machine, configure all 7 agents for your business, and support you

Not sure which one? Start with a $97 Workspace audit. Thirty minutes, actionable recommendations, and the $97 applies as credit if you go with any MojoBrain tier.

> Currently onboarding 3 clients per month.
> Book a free discovery call: calendar.app.google/35PXh7ZMU7BthPPV9

About

I'm Eysh. I spent 6 years as a Sr. Account Manager for Google Workspace at MarketStar — which means I spent 6 years watching small businesses struggle with the same operational chaos, over and over.

When I went out on my own, I built the system I wish my clients had always had. MojoBrain runs in production daily, managing my own business operations. Seven agents, one machine, zero downtime. I sell the same system I depend on.

If something in this guide resonated — let's talk.

Website	eyshtech.vip
MojoBrain	eyshtech.vip/mojobrain
Email	admin@eyshtech.vip
Book a Call	calendar.app.google/35PXh7ZMU7BthPPV9
$97 Audit	buy.stripe.com/6oU14n2xY2kR5f06Xp6AM09
LinkedIn	linkedin.com/in/eysha-penrod